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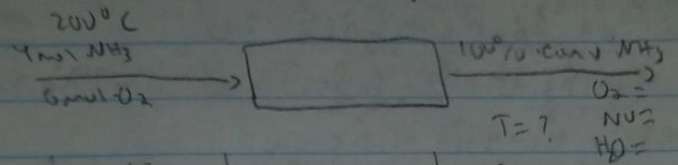
210 HW #12

1) Oxidation of Ammonia

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$$



Species	n_{in} $\frac{mol}{s}$	$\hat{H}_{in}$ $\frac{kJ}{mol}$	n_{out} $\frac{mol}{s}$	$\Delta\hat{H}_{out}$ kJ/mol
NH_3 (g)	4	$\hat{H}_1$	—	—
O_2 (g)	6	$\hat{H}_2$	$\dot{n}_3 = 1$	$\hat{H}_3$
NO (g)	—	—	$\dot{n}_4 = 4$	$\hat{H}_4$
H_2O (g)	—	—	$\dot{n}_5 = 6$	$\hat{H}_5$

Material Balance

100% conv. $\cancel{NH_3}_{out} = NH_{3\,in} - 4\dot{\xi}$ $\quad 4 - 4\dot{\xi} = 0$ $\quad \boxed{\dot{\xi} = 1\ mol/s}$

$O_{2\,out} = O_{2\,in} - 5\dot{\xi} \rightarrow 6 - 5(1) = \boxed{1 = \dot{n}_3}$

$NO_{out} = \cancel{NO_{in}}^{0} + 4\dot{\xi} \rightarrow 0 + 4(1) = \boxed{4 = \dot{n}_4}$

$H_2O_{out} = \cancel{H_2O_{in}}^{0} + 6\dot{\xi} \rightarrow 0 + 6(1) = \boxed{6 = \dot{n}_5}$

$$\hat{H}_1 = \int_{298}^{473} 27.31 + .02383t + 1.707\cdot10^{-5}t^2 - 1.185\cdot10^{-8}t^3 = \boxed{6.714\ kJ/mole}$$

$$\hat{H}_2 = \int_{298}^{473} 28.11 - 3.68\cdot10^{-6}t + 1.746\cdot10^{-5}t^2 - 1.065\cdot10^{-8}t^3 = 5.2688$$

$$\hat{H}_3 = \int_{298}^{T} 28.11 - 3.68\cdot10^{-6}T + 1.746\cdot10^{-6}T^2 - 1.065\cdot10^{-8}T^3 - 8814\ \frac{J}{mol}$$

$$\hat{H}_4 = \int_{298}^{T} 29.35 - 9.378\cdot10^{-4} + 9.747\cdot10^{-6} - 4.1878\cdot10^{-7} - 8782\ \frac{J}{mol}$$

$$\hat{H}_5 = \int_{298}^{T} 32.24 + .001924 + 1.055\cdot10^{-5} - 3.596\cdot10^{-8} - 9778.92\ \frac{J}{mol}$$

Energy Balance $\Delta H = 0$ $\quad \sum \dot{n}_e\hat{H}_e - \sum \dot{n}_i\hat{H}_i + \dot{\xi}\Delta\hat{H}^0_r$

	$\Delta\hat{H}_f$	$\Delta\hat{H}^0_c$
NH_3	-46.15	-316.8
O_2	—	—
NO	90.25	-90.2

$\Delta \hat{H}^\circ_r = 4(90.25) + 6(-241.83) - 4(-46.15) - 5(0) = -905 \frac{kJ}{mol}$

$\Delta \hat{H}^\circ_c = -4(-90.1) - 6(0) + 4(-316.8) + 5(0) = 906.4$ kJ/mol

$\Delta H = 0$ so $\xi \Delta \hat{H}^\circ_r + \sum_e \dot{n}_e \hat{H}_e - \sum_i \dot{n}_i \hat{H}_i = 0$

$(1)(-905) + [(1)\hat{H}_3 + (4)\hat{H}_4 + (6)\hat{H}_5] - [(4)(6.714) + (6)(5.2688)]$

$-905 - 58.4688 + \hat{H}_3 + 4\hat{H}_4 + 6\hat{H}_5 = 0$

$963 = \left[\dfrac{\left(28.11T + \frac{-3.68\cdot 10^{-6}}{2}T^2 + \frac{1.74\cdot 10^{-5}T^3}{3} - \frac{1.065\cdot 10^{-8}T^4}{4} - 8514\right)}{1000}\right]$ kJ/mol

$+ \left[\dfrac{4\left(29.35T - \frac{9.478\cdot 10^{-4}T^2}{2} + \frac{9.747\cdot 10^{-6}}{3}T^3 + \frac{-4.4878\cdot 10^{-9}T^4}{4} - 8782\right)}{1000}\right]$ kJ/mol

$+ \left[\dfrac{6\left(32.24T + \frac{.001924T^2}{2} + \frac{1.055\cdot 10^{-5}T^3}{3} - \frac{3.596\cdot 10^{-9}T^4}{4} - 9779\right)}{1000}\right]$ kJ/mol

Add all these together for 1 equation

$963 = \dfrac{\left(339.25T + 3.874\cdot 10^{-3}T^2 + 3.992\cdot 10^{-5}T^3 - 1.2\cdot 10^{-8}T^4 - 27075\right)}{1000}$

$990 = .339T + 3.874\cdot 10^{-6}T^2 + 3.992\cdot 10^{-8}T^3 - 1.2\cdot 10^{-11}T^4$

$T = 2657$ K or -3626 K

Using NIST Shomate equation

$\hat{H}_1 = \int_{.298}^{.473} C_{p\,NH_3} = H^\circ - H^\circ_{298} = At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} + \frac{E}{t} + F - H$

$\hat{H}_1 = 6.69866$, $\hat{H}_2 = 5.24836$, $\hat{H}_3 = H^\circ - H^\circ_{298}$

$\hat{H}_3 = 31.32234t - \frac{20.23531t^2}{2} + \frac{57.87649t^3}{3} - \frac{36.50624t^4}{4} + \frac{0.007374}{t} - 8.903$

$\hat{H}_4 = 23.83491t + \frac{12.58878t^2}{2} + \frac{-1.139T^3}{3} - \frac{1.49746T^4}{4} - \frac{0.214194}{t} - 6.932$

$\hat{H}_5 = 30.0920T + \frac{6.832514T^2}{2} + \frac{6.793435t^3}{3} + \frac{-2.53448T^4}{4} - \frac{0.082139}{t} - 9.0546$

$\Delta H = 0 = \xi \Delta \hat{H}^\circ_r + \sum_e \dot{n}_e \hat{H}_e - \sum_i \dot{n}_i \hat{H}_i$ $\quad \xi = 1$ mols $\quad \Delta \hat{H}^\circ_r = -905.4$ kJ/mol

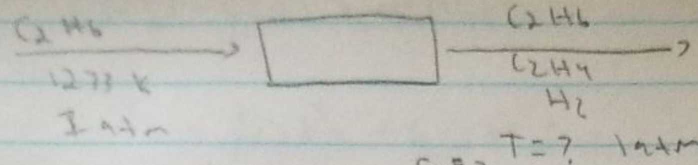
$\Delta H = 0 = -905.4(1) + \hat{H}_3 + 4\hat{H}_4 + 6\hat{H}_5 - 4(6.69866) + 6(5.24836) - 58.2848$

plug in $\hat{H}_3, \hat{H}_4, \hat{H}_5$ in solver to set equation:

$1054.6274 = 307.214t + 35.55738t^2 + 31.357t^3 - 14.4t^4 - \frac{1.2758}{t}$

you find: $t = 2769.66$ K

2. $C_2H_6(g) \longleftrightarrow C_2H_4(g) + H_2(g)$ basis: $\dot{n}_{C_2H_6} = 1$ mol



$$K_p = \frac{x_{C_2H_4}\,x_{H_2}}{x_{C_2H_6}} P = K_0 \cdot e^{\left[-\frac{E}{T}\right]}$$

$K_0 = 7.28 \cdot 10^6$ atm

$E = 17{,}000$ K

A) Find conversion $f_{C_2H_6} = f = \sqrt{\frac{K_p}{P_{atm} + K_p}} = \frac{\text{moles } C_2H_6 \text{ reacted}}{\text{moles } C_2H_6 \text{ fed}}$

$\dot{n}_{C_2H_6,out} = \dot{n}_{C_2H_6,in} - \xi \rightarrow 1 - \xi = 1 - f$ $\quad f = \frac{\xi}{n_{C_2H_6,in}} = \frac{\xi}{1} = \boxed{\xi}$

$\dot{n}_{C_2H_4,out} = \cancel{n_{C_2H_4,in}} + \xi = f$

$\dot{n}_{H_2,out} = \dot{n}_{H_2,in} + \xi = f$ $\quad K_p = \left[\frac{f \cdot f}{(1-f)(1+f)}\right] = \left[\frac{f^2}{1-f^2}\right] P$

$n_{TOT} = 1 + \xi$

$K_p(1-f^2) = f^2 P$

$K_p - K_p f^2 = f^2 P \rightarrow K_p = f^2 P + K_p f^2 \rightarrow K_p = f^2 (P + K_p)$

$\frac{K_p}{P + K_p} = f^2$ $\quad \boxed{f = \sqrt{\frac{K_p}{P + K_p}}}$

$f = \frac{1}{1 + \phi T}$ $\quad \phi T = \dfrac{\Delta \hat{H}_r(@1273\,K) - \int_T^{1273} [C_{P\,C_2H_4} + C_{P\,H_2}]\,dT}{\int_T^{1273} [C_{P\,C_2H_6}]\,dT}$

$$= \frac{145.1 - \left\{\left[9.419T + \frac{.1147T^2}{2}\right]_T^{1273} + \left[26.90T + \frac{.004167T^2}{2}\right]_T^{1273}\right\}}{\left[11.35T + \frac{.1392T^2}{2}\right]_T^{1273}}$$

$$= \frac{145.1 - [11990.387 + 92937.338 - 7.419T - 0.05735T^2 + 34243.7 + 3376.372 - 26.9T + .002083T^2]}{[14448.55 + 112788.81 - 11.35T + 0.0686T^2]}$$

$$\phi T = \frac{145.1 - [142547.795 - 36.319T - .05527T^2]}{[127237.36 - 11.35T - 0.0696T^2]}$$

$\psi(T) = 0 = \dfrac{1}{1 + \text{''} \quad \text{''}}$

$K_p(T) = 7.28 \cdot 10^6 e^{\left(\frac{-13{,}000}{T}\right)}$

$$\frac{1}{1+\phi T} = f = \sqrt{\frac{K_p}{1+K_p}} = \frac{1}{1 + \dfrac{145.1 - [142547.795 - 36.32T - .05527T^2]}{[127237.36 - 36.319T - 0.05527T^2]}}$$

$e = 0.01, 0.05, 0.1, 0.5, 1, 5$

Comment Summary

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1. -5
   For not having plots

Page 3

2. -6 for not having plots